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BELGIAN CO-ORDINATED COLLECTIONS OF MICRO-ORGANISMS
SECTION MYCOLOGY AND AEROBIOLOGY

 **sciensano**

GREAT AT SMALL THINGS

Expanding environmental surveillance of azole-resistant *A. fumigatus* in Belgium

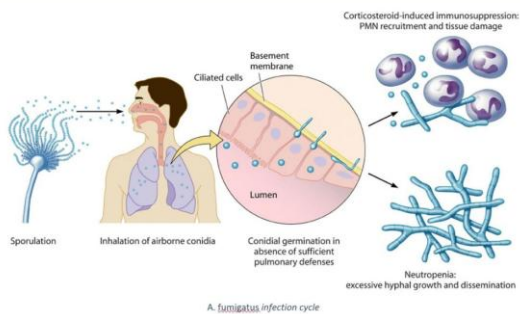
Hanne Debergh, PhD

12/06/2025

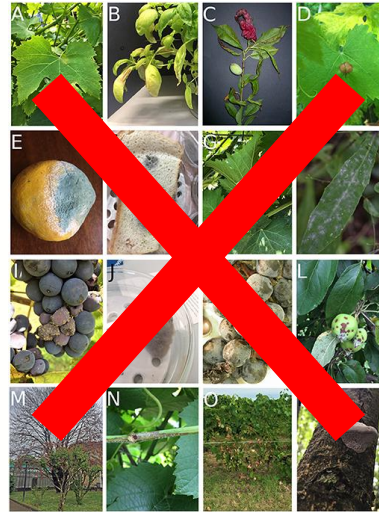
Belgian
Science
Policy
Office



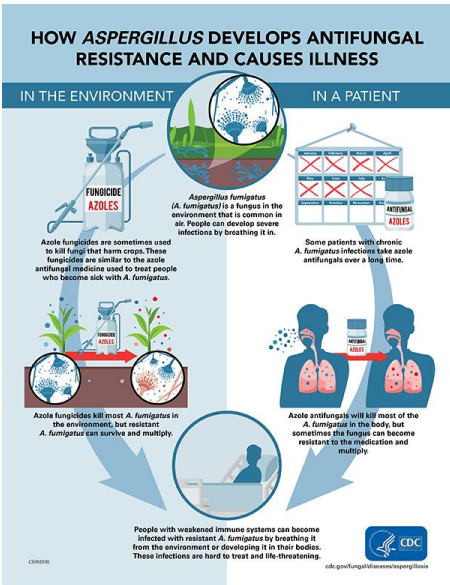
ENVIRONMENTAL SURVEILLANCE?



Source : [T. Dagenais et al., 2009](#)



ENVIRONMENTAL SURVEILLANCE?



WHERE ARE WE AT?



- What has already been done in Belgium?
- What does the future hold?

TECHNICAL REPORT

Risk assessment on the impact of environmental usage of triazoles on the development and spread of resistance to medical triazoles in *Aspergillus* species

Recommendations

- Improve epidemiological surveillance through:
 - Routine triazole susceptibility testing for clinical isolates (if antifungal treatment is indicated);
 - Increased and continuous surveillance of triazole resistance in *A. fumigatus* in each EU Member State;
 - Strengthen mycology function at ECDC to monitor disease frequency and triazole resistance, facilitate Member State adoption of recommendations and impact of any public health interventions.
- Develop molecular methods to detect triazole resistance in culture-negative specimens and implement them in laboratory practice.
- Investigate the environmental origin through environmental and laboratory studies, including:
 - Field studies with different triazoles which are putative resistance generators with quantitative and sensitive means of detecting triazole resistance in the environment;
 - Extensive and continued environmental studies;
 - Non-agricultural studies (triazoles are also used extensively by industry, application professionals and amateurs for disease control of lawns/ornamentals and preservation of materials such as wood, coating/paints and plasters);
 - Studies on the reversal of rising resistance rates by restricting certain triazole applications will help to identify appropriate public health measures.

Table 3. WHO fungal priority pathogens list

Critical group	High group	Medium group
 <i>Cryptococcus neoformans</i>	 <i>Nakaseomyces glabrata</i> (<i>Candida glabrata</i>)	 <i>Scedosporium</i> spp.
 <i>Candida auris</i>	 <i>Histoplasma</i> spp.	 <i>Lomentospora prolificans</i>
 <i>Aspergillus fumigatus</i>	 Eumycetozoa causative agents	 <i>Coccidioides</i> spp.
 <i>Candida albicans</i>	 Mucorales	 <i>Pichia kudriavzevii</i> (<i>Candida krusei</i>)
	 <i>Fusarium</i> spp.	 <i>Cryptococcus gattii</i>
	 <i>Candida tropicalis</i>	 <i>Talaromyces marneffei</i>
	 <i>Candida parapsilosis</i>	 <i>Pneumocystis jirovecii</i>
		 <i>Paracoccidioides</i> spp.

Approved: 13 December 2024

DOI: 10.2903/j.efsa.2025.9200

SCIENTIFIC REPORT



Impact of the use of azole fungicides, other than as human medicines, on the development of azole-resistant *Aspergillus* spp.

European Food Safety Authority (EFSA) |

European Centre for Disease Prevention and Control (ECDC) | European Chemicals Agency

(ECHA) | European Environment Agency (EEA) | European Medicines Agency (EMA) |

European Commission's Joint Research Centre (JRC)*



Impact of the use of azole fungicides, other than as human medicines, on the development of azole-resistant *Aspergillus* spp.

- Sales of azoles in EU/EEA 2010 – 2021: $\pm 120\,000\text{t}$ for non-medical purposes
>119 000t used for plant protection (annual average of 10 00t)



- Evidence supports a link between environmental [azole fungicide exposure](#), azole resistance selection in *Aspergillus* spp. and [cross-resistance to medical azoles](#). Further research is required to estimate the extent of the contribution on infections caused by azole-resistant *A. fumigatus*.



Impact of the use of azole fungicides, other than as human medicines, on the development of azole-resistant *Aspergillus* spp.

- Based on ecological selection dynamics, identified **environmental hotspots** for selection of azole resistance include:
 - **agriculture**: stockpiled agricultural waste used to improve and fertilise soil for various crops (indoor fruiting vegetables, wine grape, corn, sugar beet, olives, pome fruit, citrus fruit and field heaps);
 - **biocides**: freshly-cut wood.



Impact of the use of azole fungicides, other than as human medicines, on the development of azole-resistant *Aspergillus* spp.

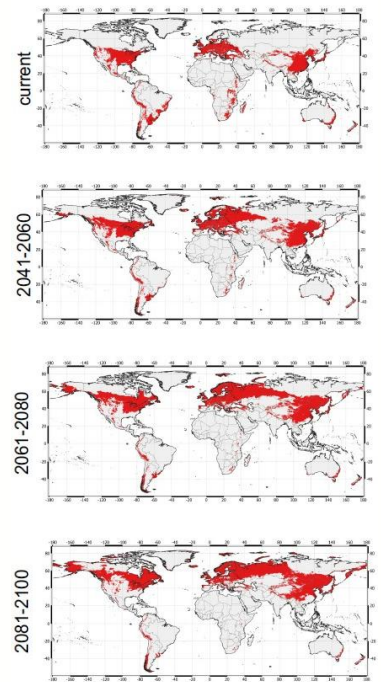
- The reported prevalence of **human azole-resistant *A. fumigatus* infections** among all *A. fumigatus* infections varies by the type of aspergillosis disease and the geographic region
 - IA: 0.7–63.6 %,
 - CPA: 5.9–59.2 %
 - APA: 2.3–42.8 %.
- Azole resistance has significant clinical implications, especially for azole-resistant *A. fumigatus* IA, with **mortality rates reaching 36–100 %**.



A. FUMIGATUS

- Climate change
 - Increase the range of currently-pathogenic species
 - Increase the thermotolerance allowing more fungal species to survive at mammalian body temperature

Aspergillus fumigatus



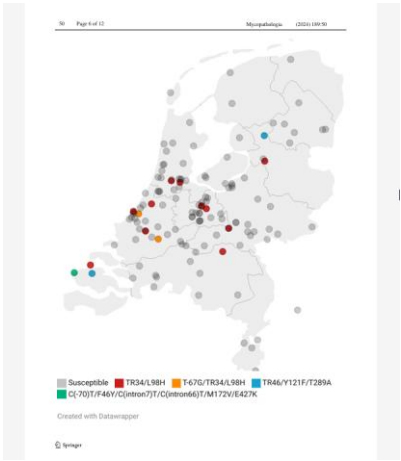
ONE HEALTH - ANIMALS

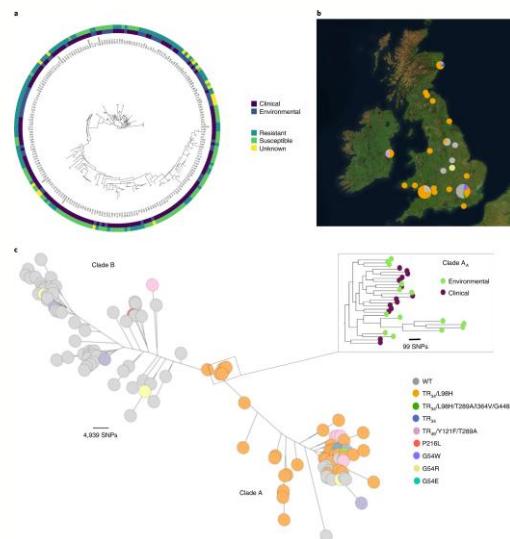
Mycopathologia (2024) 189:50
<https://doi.org/10.1007/s11046-024-00850-5>

ORIGINAL ARTICLE

Azole Resistance in Veterinary Clinical *Aspergillus fumigatus* Isolates in the Netherlands

Marloes A. M. van Dijk · Jochem B. Buil ·
Marlou Tehupeiory-Kooreman · Marian J. Broekhuizen · Els M. Broens ·
Jaap A. Wagenaar · Paul E. Verweij





Rhodes, J. *et al.* Population genomics confirms acquisition of drug-resistant *Aspergillus fumigatus* infection by humans from the environment. *Nat Microbiol* 7, 663–674 (2022).

HOTSPOTS ?

Environmental Hotspots for Azole Resistance Selection of *Aspergillus fumigatus*, the Netherlands

Sijmen E. Schoustra, Alfons J.M. Debets, Antonius J.M.M. Rijs,¹ Jianhua Zhang, Eveline Snelders, Peter C. Leendertse, Willem J.G. Melchers, Anton G. Rietveld, Bas J. Zwaan, Paul E. Verweij

RESEARCH ARTICLE

Detection of azole-resistant *Aspergillus fumigatus* in the environment from air, plant debris, compost, and soil

Julen E. Gómez-Londoño¹, Marin T. Brewer²*

Azole-resistant
Identifying key
resistance

Caroline Burks, Alexandria D.

Belgium?

ment:
al

Citizen Science Surveillance of Triazole-Resistant *Aspergillus fumigatus* in United Kingdom Residential Garden Soils

Jennifer M. G. Shelton,^{a,b} Roseanna Collins,^c Christopher B. Uzzell,^a Asmaa Alghamdi,^d Paul S. Dyer,^d Andrew C. Singer,^b Matthew C. Fisher^a

The Multi-Fungicide Resistance Status of *Aspergillus fumigatus* Populations in Arable Soils and the Wider European Environment

Bart Fraaije^{1,2*}, Sarah Atkins², Steve Hanley², Andy Macdonald² and John Lucas²



SURVEILLANCE

Doubling of triazole resistance rates in invasive aspergillosis over a 10-year period, Belgium, 1 April 2022 to 31 March 2023

Lize Cuypers^{1,2,*}, Robina Aerts^{2,3,*}, Otto Van de gaer¹, Lore Vinken², Rita Merckx², Veerle Gerils¹, Greetje Vande Velde⁴, Agustin Reséndiz-Sharpe⁴, Johan Maertens^{2,5}, Katrien Lagrou^{1,2}

1. Department of Laboratory Medicine, National Reference Centre for Mycosis, University Hospitals Leuven, Leuven, Belgium

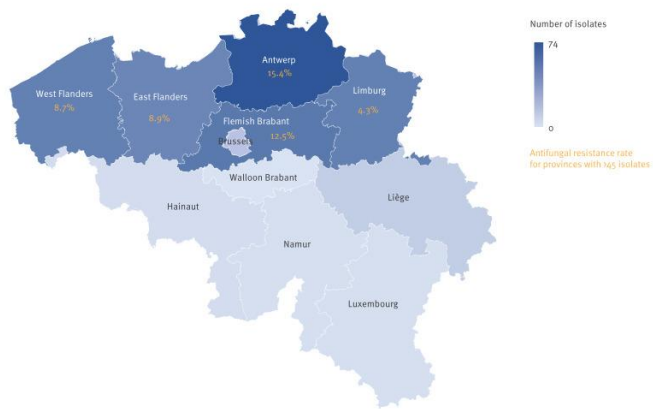
2. Department of Microbiology, Immunology and Transplantation, Laboratory of Clinical Microbiology, KU Leuven, Leuven, Belgium

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4. Department of Imaging and Pathology, Biomedical MRI unit, KU Leuven, Leuven, Belgium

5. Department of Haematology, University Hospitals Leuven, Leuven, Belgium

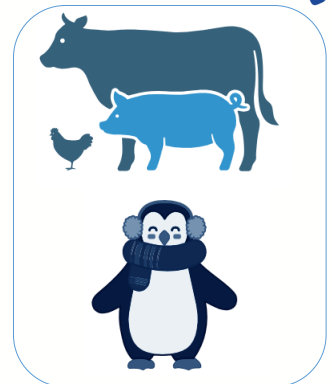
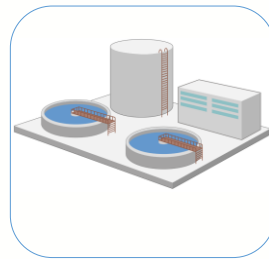
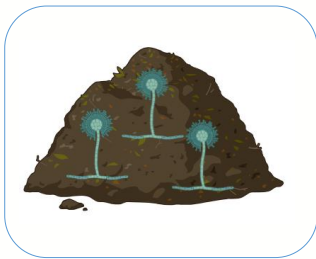
* These authors contributed equally to this work and share first authorship.



WHAT HAS ALREADY BEEN DONE IN BELGIUM?



WHAT HAS ALREADY BEEN DONE IN BELGIUM?



ENVIRONMENTAL SURVEILLANCE IN BELGIUM

Medical Mycology 2024, 62, myx055
<https://doi.org/10.1093/mmy/myx055>
Original Article
Advance access publication date: 20 May 2024



Detection of pan-azole resistant *Aspergillus fumigatus* in horticulture and a composting facility in Belgium

Hanne Debergh^{1,2,*}, Philippe Castelain^{3,2}, Karine Goens¹, Paulien Lefevère¹, Elien De Vits¹, Marc Visser⁴, Liesbet Blindeman⁴, Charlotte Bataille⁵, Claude Ann Packeu^{1,6}

Table 1 - Environmental sampling locations and sampling approaches

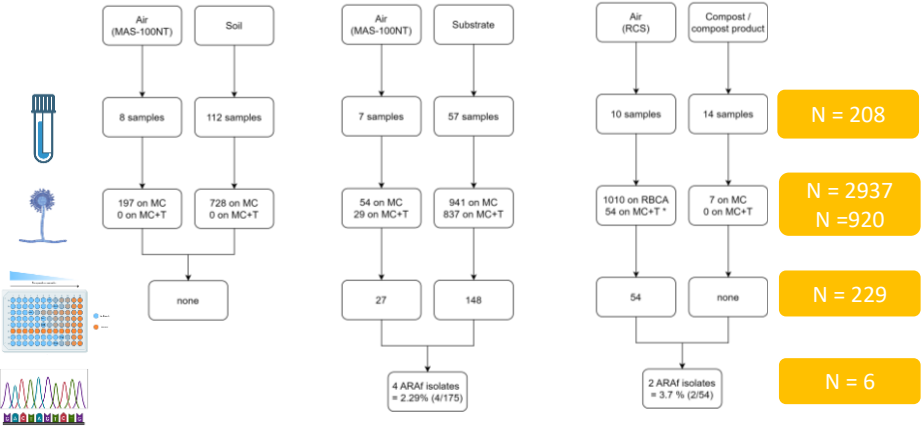
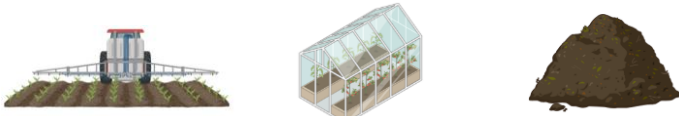
Environmental sampling locations		Sampling dates	Sampling approach	Air impaction method
Compost	Facility 1	August 2020	Air and compost	RCS ⁶
	Facility 2	April 2021	Air and compost	RCS ⁶
Agriculture	Wheat	May – July 2022 ¹	Air and soil	MAS-100NT ⁶
Horticulture	Roses	June 2020 ²	Air and substrate	MAS-100NT ⁶
	Hibiscus	June 2020 ³	Air and substrate	MAS-100NT ⁶
	Primula	January 2021 ⁴	Air and substrate	MAS-100NT ⁶
	Residue heap ⁵	June 2020	Soil	NA

¹ four sampling campaigns were performed: before treatment (11/05/2022), 6 days after treatment (18/05/2022), 3 weeks after treatment (2/06/2022) and during harvest (28/07/2022); ² sampling was performed 3 days after last treatment; ³ sampling was performed 28 days after last treatment; ⁴ sampling was performed 41 days after last treatment; ⁵ The residue heap acts as a bioremediation location where left-overs of plant protection product (PPP) spray dilutions are poured in order to eliminate PPP waste; MAS-100NT⁶ = Microbial Air Sampler; RCS⁶ = Ruter Centrifugal Sampler; NA = not applicable.



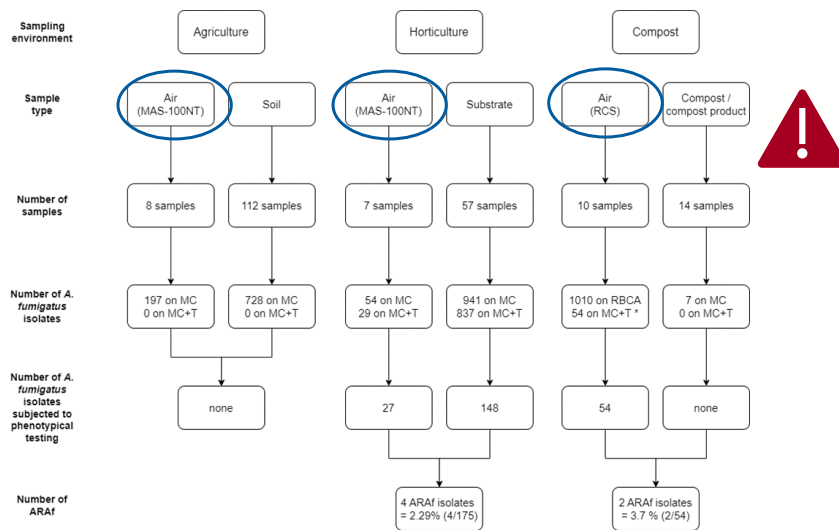


STUDY 3 – ARAf in the environment



Introduction
Objectives
Discussion
EXPERIMENTS

ENVIRONMENTAL SURVEILLANCE IN BELGIUM



International standardization is necessary!

Limitations:
Contamination
Disproportion in samples

Table 4 – Minimal inhibitory concentration results and *cyp51A* mechanism

Sampling site	Sample type	Accession nr*	MIC (mg/L) ¹				<i>cyp51A</i> mutations
			VOR	ITC	ISA	POSA	
Compost (F1)	Air	28546	8	>16	8	1	TR34/L98H
	Air	28547	4	>16	8	1	TR34/L98H
Horticulture	Air (hibiscus)	28954	8	8	2	0,5	TR34/L98H
Horticulture	Soil (residue heap)	28947	>16	0,5	>64	0,5	TR46/Y121F/T289A
Horticulture	Substrate (Primula)	28553	4	>16	8	1	TR34/L98H
Horticulture	Substrate (Primula)	28554	>16	4	8	2	No mutation detected

MIC = Minimal inhibitory concentration; VOR = Voriconazole; ITC = Itraconazole; ISA = Isavuconazole; POSA = Posaconazole; F1 = Facility 1; ¹ MIC determined using the EUCAST guidelines ⁵⁶; * The isolates were deposited in the BCCM/IHEM collection under the mentioned accession numbers (<https://bccm.belspo.be/about-us/bccm-ihem>).

Prevalence of 2,62% ARAf
(6/229)

Presence of resistant strains in

horticulture and compost
called "hotspots" defined as an environment in which: (1) the physical, biotic and abiotic conditions facilitate the growth of the fungus and from which the fungus can spread; (2) this growth can take place for prolonged periods and the fungus can complete all the stages of its growth cycle; and (3) azoles are present, in different concentrations sufficient to select in populations, and combinations (Schoustra et al., 2019) (Fig. 1). Studies that sampled environments for the presence

concentrations + more systemic
sampling

→ NAP AMR 2025-2028



GLOBAL ROADMAP ON ANTIFUNGAL RESISTANCE

➔ a strategic framework to guide global action, inform policy, and stimulate innovation in the fight against antifungal



SURVEILLANCE IN ANIMALS



Pan-azole resistance in clinical *Aspergillus fumigatus* isolates carrying TR34/L98H from birds and mammals in Belgium

Hanne Debergh^{a,b,*}, Roel Haesendonck^c, Nadine Botteldoorn^d, An Martel^e, Frank Pasmans^e, Claude Saegerman^b, Ann Packer^{a,f}

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SURVEILLANCE IN ANIMALS – BIRDS AND MAMMALS

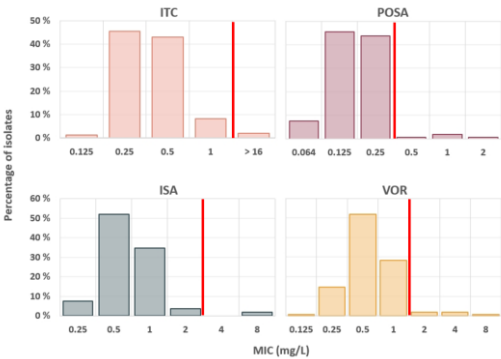
01/2020 – 01/2024

Taxonomic order		Origin				Sample type				Necropsy
		Pet	Production	Wild	Zoo	Digestive system	Reproduction system	Respiratory system	Other*	
Avian (n = 71)	Accipitriformes (n = 1)	0	0	0	1	0	0	1	0	1
	Anseriformes (n = 4)	1	0	0	3	0	0	3	1	3
	Charadriiformes (n = 12)	0	0	12	0	0	0	12	0	12
	Columbiformes (n = 30)	29	0	1	0	0	0	27	3	30
	Galliformes (n = 3)	2	1	0	0	0	0	2	1	3
	Gaviiformes (n = 1)	0	0	1	0	0	0	1	0	1
	Passeriformes (n = 6)	2	0	2	2	2	0	4	0	6
	Pelecaniformes (n = 1)	0	0	0	1	0	0	1	0	1
	Podicipediformes (n = 1)	0	0	1	0	0	0	1	0	1
	Psittaciformes (n = 7)	6	0	0	1	0	0	7	0	7
	Sphenisciformes (n = 4)	0	0	0	4	0	0	4	0	3
	Suliformes (n = 1)	0	0	1	0	0	0	1	0	0
	Struthioniformes (n = 1)	0	0	0	0	0	0	0	0	0
Mammal (n= 81)	Artiodactyla (n = 61)	0	61	0	0	48	3	9	1	60
	Carnivora (n = 7)	7	0	0	0	0	0	5	2	1
	Perissodactyla (n = 13)	13	0	0	0	0	3	10	0	2
Total (n = 152)		60	62	18	12	50	6	88	8*	131
		(39.5 %)	(40.8 %)	(11.8 %)	(7.9 %)	(32.9 %)	(3.9 %)	(57.9 %)	(5.3 %)	(86.2%)

* other sample types include: abscess, ear, skin, kidney, pericardium swab, ear canal, tongue swab and foot abscess.

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SURVEILLANCE IN ANIMALS – BIRDS AND MAMMALS



3.3% overall resistance rate

Accession number	Species	Sample type	Year of isolation	MIC (mg/L) ¹				Cyp51A mutations
				VOR	ITC	ISA	POS	
28429	Cow	Digestive	2020	2	0,5	2	0,5	F46Y, M172V, E427K, N248T, D255E
28552	Cat	Ear	2021	8	1	2	2	no mutation
28550	Cow	Digestive	2020	4	>16	8	1	TR34/L98H
29030	Pigeon	Respiratory	2023	4	>16	8	1	TR34/L98H
29031	Pigeon	Respiratory	2023	4	>16	8	1	TR34/L98H

SURVEILLANCE IN ANIMALS



antibiotics



Article

Pulmonary Aspergillosis in Humboldt Penguins—Susceptibility Patterns and Molecular Epidemiology of Clinical and Environmental *Aspergillus fumigatus* Isolates from a Belgian Zoo, 2017–2022

Hanne Debergh ^{1,2} , Pierre Becker ³, Francis Vercammen ⁴ , Katrien Lagrou ^{5,6} , Roel Haesendonck ⁷, Claude Saegerman ^{2,*}  and Ann Packeu ^{1,3}



SURVEILLANCE IN ANIMALS – PENGUINS



High rate of avian aspergillosis in a Belgian zoo

Measures taken by the zoo

Environmental sampling in penguin housing

Laboratory analyses (environment + clinical)

Transmission between environment and animal?

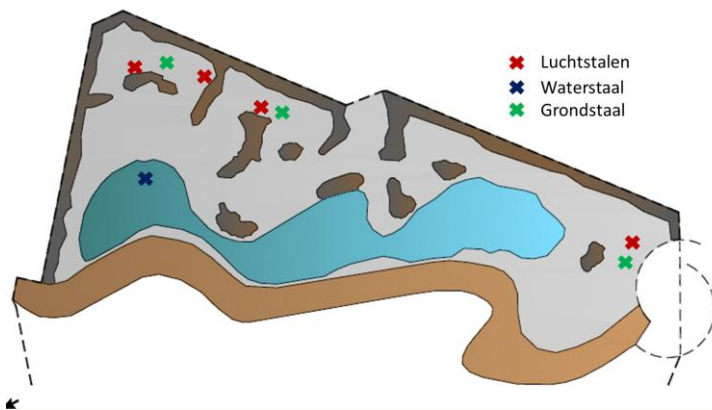
SURVEILLANCE IN ANIMALS – PENGUINS



	Living animals	Dead animals	Death < 2 months old	Autopsies	Aspergillosis	Deathrate (%)	Aspergillosis (%)
2017	99	27	11	15	13	27.3	48.1
2018	78	37	3	34	26	47.4	70.3
2019	41	2	0	2	1	4.9	50.0
2020	45	6	2	4	1	13.3	16.7
2021	52	20	4	16	7	38.5	35.0

27 *A. fumigatus*
isolates in our
collection

SURVEILLANCE IN ANIMALS – PENGUINS



- ✕ Luchtstaal
- ✕ Waterstaal
- ✕ Grondstaal

Season	Sampling date	# birds
Spring	1 st of April 2021	41
Summer	29 June 2021	43
Fall	13 October 2021	28
Winter	12 January 2022	28

SURVEILLANCE IN ANIMALS – PENGUINS



Table 2. Demography of the bird specimens included in this study, and minimum inhibitory concentration towards antifungal drugs of the *A. fumigatus* strains isolated from the autopsied bodies.

IDA. <i>Fumigatus</i> Strain	VOR	ITC	ISA	POSA	Host Species	Date of Death	Age	Treatment Since **
21-0659	1	0.5	0.5	0.125	Humboldt penguin	28/12/2017	16y10m5d	N/A
21-0660	1	0.25	1	0.25	Humboldt penguin	29/12/2017	20y7d	N/A
21-0029	0.25	0.25	0.25	0.125	Humboldt penguin	12/11/2018	11y5m18d	N/A
21-0030	0.5	0.25	0.5	0.125	Humboldt penguin	12/12/2018	11y5m18d	N/A
21-0661	1	0.5	1	0.25	Humboldt penguin	1/01/2018	20y9m23d	N/A
21-0662	4 *	>16 *	4 *	0.5 *	Humboldt penguin	26/02/2018	8y9m27d	16/02/2017
21-0663	1	0.5	0.5	0.25	Humboldt penguin	2/03/2018	9m7d	N/A
21-0664	0.5	0.25	0.5	0.25	Humboldt penguin	29/03/2018	19y17d	N/A
21-0665	2	0.5	1	0.25	Crested partridge	4/04/2018	5m15d	N/A
21-0666	0.5	0.25	0.5	0.125	Humboldt penguin	7/04/2018	13y11m23d	N/A
21-0667	1	0.5	0.5	0.25	Humboldt penguin	9/05/2018	14y1m2d	2017 †
21-0668	1	0.5	0.5	0.25	Humboldt penguin	18/07/2018	11y1m17d	N/A
21-0669	0.5	0.25	0.5	0.064	Humboldt penguin	19/07/2018	11y1m24d	N/A
21-0670	1	0.55	1	0.25	Humboldt penguin	25/07/2018	11y2m	N/A
21-0671	0.5	0.25	0.5	0.25	Humboldt penguin	7/08/2018	2y1m6d	N/A
21-0672	0.5	0.5	0.5	0.25	Humboldt penguin	9/08/2018	2y1m8d	N/A
21-0673	0.5	0.5	0.5	0.125	Humboldt penguin	8/08/2018	12y1m24d	N/A
21-0674	0.5	0.5	2	0.25	Humboldt penguin	17/08/2018	7y4m7d	N/A
21-0675	1	0.5	2	0.25	Humboldt penguin	5/09/2018	17y4m15d	N/A
21-0676	1	1	1	0.5 *	Humboldt penguin	5/09/2018	7y3m11d	N/A
21-0677	0.5	0.5	1	0.25	Humboldt penguin	4/09/2018	3y4m13d	N/A
21-0678 *	0.5	0.5	1	0.5 *	Humboldt penguin	13/09/2018	9y5m1d	N/A
21-0679	1	1	2	0.25	Humboldt penguin	22/09/2018	22y5m1d	16/2/2017
21-0680	1	1	1	0.5 *	Humboldt penguin	28/09/2018	17y5m8d	2018 †
21-0034	0.5	0.25	0.5	0.125	Red-billed blue naggie	17/11/2019	4m, 4d	N/A
21-0035	0.25	0.25	0.25	0.125	Chilean flamingo	22/02/2020	45y7m19d	N/A
21-0036	0.5	0.25	0.5	0.125	Chestnut-backed thrush	8/10/2020	6y1m5d	N/A
21-0039	0.5	0.25	0.5	0.125	Crested cormorant	5/03/2021	6y11m21d	N/A
21-0428/21-0488	0.25	0.25	0.25	0.125	Humboldt penguin	26/05/2021	12y1m17d	2020 †
21-0494	2	1	2	0.5 *	Humboldt penguin	1/08/2021	4y3m3d	N/A
21-0523	0.5	0.25	0.5	0.125	Humboldt penguin	6/09/2021	20y7m23d	N/A
21-0524	4 *	2	4 *	0.5 *	Humboldt penguin	1/08/2021	5y3m12d	N/A
21-0525	0.5	0.5	0.5	0.125	African penguin	12/08/2021	11y3m2d	N/A
22-0592	0.25	0.25	0.5	0.032	Humboldt penguin	29/08/2022	22y4m12d	N/A

N/A = not applicable, VOR = voriconazole, ITC = itraconazole, ISA = isavuconazole, and POSA = posaconazole; * indicates resistance following the EUCAST clinical breakpoints for fungi, v10.0 [39]; ** starting date of treatment with 20 mg/kg itraconazole daily until death; † exact starting date of treatment is unknown.

SURVEILLANCE IN ANIMALS – PENGUINS



Table 3. Antifungal susceptibility testing results of the isolates displaying antifungal resistance against at least one medical azole, and associated *cyp51A* mutation.

Source	ID Strain	Date of Isolation	VOR	ITC	ISA	POSA	<i>cyp51A</i> Mutation
Environmental	21-0468	29/06/2021	2 *	2 *	4 *	0.5 *	TR34/L98H
	21-0503	13/10/2021	4 *	>16 *	4 *	0.5 *	TR34/L98H, G54R
	21-0506	13/10/2021	4 *	>16 *	4 *	0.5 *	TR34/L98H
	21-0510	13/10/2021	4 *	>16 *	16 *	2 *	TR34/L98H
	21-0515	13/10/2021	2 *	2 *	4 *	0.5 *	TR34/L98H
	21-0516	13/10/2021	2 *	2 *	4 *	0.5 *	TR34/L98H
	21-0517	13/10/2021	2 *	2 *	4 *	0.5 *	TR34/L98H
	21-0518	13/10/2021	2 *	2 *	4 *	0.5 *	TR34/L98H
Veterinary	21-0662	26/02/2018	4 *	>16 *	4 *	0.5 *	TR34/L98H
	21-0676	5/09/2018	1	1	1	0.5 *	no known mutations found
	21-0678	13/09/2018	0.5	0.5	1	0.5 *	no known mutations found
	21-0680	28/09/2018	1	1	1	0.5 *	F46Y, M172V, E427K
	21-0494	1/08/2021	2 *	1	2 *	0.5 *	no known mutations found
	21-0524	1/08/2021	4 *	2 *	4 *	0.5 *	TR34/L98H

* phenotypic resistance according to the EUCAST clinical breakpoints (v10.0) [39].

SURVEILLANCE IN ANIMALS – PENGUINS

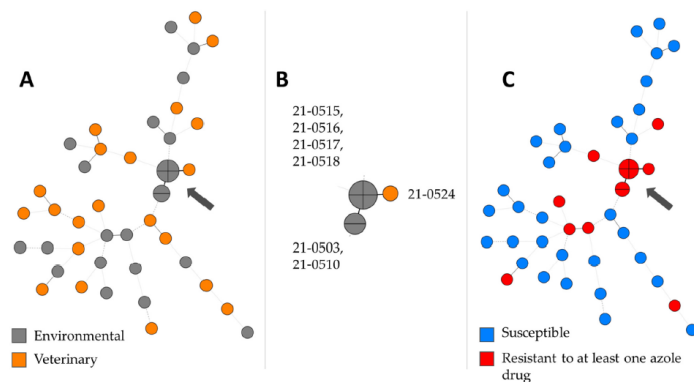
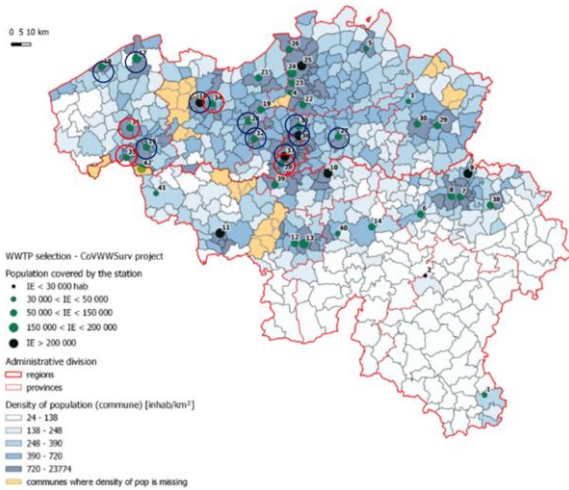


Figure 3. Minimum spanning tree displaying the 41 different genotypes obtained from 45 *Aspergillus fumigatus* isolates, based on 9 variable-number tandem-repeat loci. The genotypes are represented by circles. The length and thickness of the connecting lines between the circles show the similarity between the profiles. (A) Mapping of the source of the isolate (i.e., isolated from the environment or from an infected animal). (B) Detailed map of the cluster indicating the strains ID. (C) Mapping of the resistance profile towards azole drugs (i.e., susceptible to all tested drugs, or resistant to at least one tested drug). The arrow shows the seven clustered isolates.

ENVIRONMENTAL SURVEILLANCE – WASTEWATER



4 seasons, 3 weeks / season
1/10 dilution
MC and MC+T, 48h at 48°C

WWTP site	Population covered by the WWTP
Brussels North	1100000
Brussels South	360000
Gent	221791
Brugge	181266
Oostende	136065
Leuven	121597
Harelbeke	108627
Grimbergen	107414
Liederkerke	101238
Aalst	98618



ENVIRONMENTAL SURVEILLANCE – WASTEWATER



Sampling round 1

Season	Colonies on MC+T	# Resistant isolates
Fall	33	4
Winter	116	3
Spring	83	4
Summer	115	3
TOTAL	347	14 (4,03%)

Sampling round 2

Season	Colonies on MC+T	# Resistant isolates
Fall	103	9
Winter	83	8
Spring	86	4
Summer	44	1
TOTAL	316	22 (6.96%)

13	TR34/L98H
2	TR46/Y121F/T289A/(G448S)
2	F46Y/M172V/(N248T/D255E)/E427K
19	ongoing



ENVIRONMENTAL SURVEILLANCE – HOSPITALS



- Flanders
- Brussels
- Wallonia

A. Fumigatus inside vs.
outside the hospital



four sampling periods

ENVIRONMENTAL SURVEILLANCE – WHAT WILL THE FUTURE BRING?

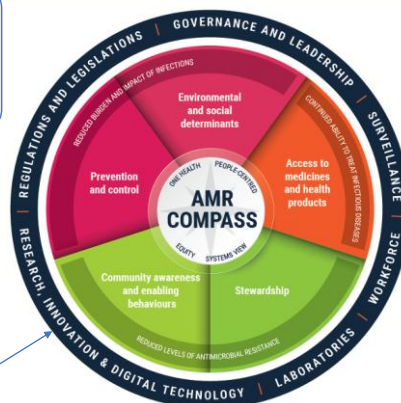


ENVIRONMENTAL SURVEILLANCE – NAP AMR



‘Development of measures to decrease the spread of resistant isolates from a WHO priority fungal pathogen in the Belgian environment’

Enabler



ENVIRONMENTAL SURVEILLANCE – NAP AMR



- Characterize **hotspots** for azole resistance development in Belgium
- Provide valuable information and data regarding the **emergence and selection** of azole resistance in *A. fumigatus*
- Support the **development of measures to decrease AMR**
- **Raise awareness** among professional organizations and workers in the field
- Establish guidelines to **decrease the occurrence and the spread of azole resistance** in *A. fumigatus* in the environment, e.g. on waste management from azole-treated crops

ABR2AFR

International One Health
Stakeholder Meeting on

Antifungal Resistance



CONCLUSIONS



- The problem of azole resistance in *A. fumigatus* is on the European agenda (joint European report)
- Potential hotspots in Belgium were identified: industrial composting facilities and horticulture
- Doubling of azole resistance in the clinics warrants more elaborate environmental surveillance, fast!
- The NAP AMR 2025-2029 will tackle several important knowledge gaps concerning azole resistance in Belgium.